

Assignment 4

Please submit the code and report (in pdf format) on Blackboard system before 23:59 June 7. Report can be written either in English or Chinese. Name your report as

`studentID_Name.pdf`

Q1 User analyses

The customer data is attached. Analyze the data to better understand customer behavior. Before analyses, you may need to observe the data and do some data cleaning.

- (1) [2 points] Use Pyspark K-means to cluster the users into groups. You can use all features or part of the features for clustering. Also, you will decide the optimal number of clusters.
- (2) [1 points] visualize the clustering with t-SNE.
- (3) [2 points] Observe the customer clusters and give your explanations to the results.

Data Attribute Information:

- InvoiceNo: Invoice number. Nominal, a 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c', it indicates a cancellation.
- StockCode: Product (item) code. Nominal, a 5-digit integral number uniquely assigned to each distinct product.
- Description: Product (item) name. Nominal.
- Quantity: The quantities of each product (item) per transaction. Numeric.
- InvoiceDate: Invoice Date and time. Numeric, the day and time when each transaction was generated.
- UnitPrice: Unit price. Numeric, Product price per unit in sterling.

- CustomerID: Customer number. Nominal, a 5-digit integral number uniquely assigned to each customer.
- Country: Country name. Nominal, the name of the country where each customer resides.

Note:

- The report contains the screenshot of the program result. You can include the explanation of each function you used in the report or in the code as comments.
- The python code can be written in jupyter or python files.

Q2 Startup Idea Evaluation

Assume you intend to launch a startup and conduct brainstorming to develop viable product concepts.

(1) [3 points] Prompt LLMs to generate 10 diverse and practical startup ideas. You can refer to this [link](#) to facilitate the idea generation process. You may run the prompt several times or use different in-context examples.

(2) [2 points] Prompt LLMs to design the system architectures for your top two preferred ideas, and generate corresponding Docker Compose YAML configuration files.